

# Lines and Planes in Three Dimensions

## ANSWER KEY



### The vector equation of a line

- 1 Write the vector equation of the line through the point  $(2, -1, 3)$  with direction vector  $(1, 2, -1)$ .  
 $\mathbf{r} = (2, -1, 3) + t(1, 2, -1)$ , where  $t \in \mathbb{R}$ .
- 2 Find the vector equation of the line through the two points  $A(1, 0, 2)$  and  $B(4, 3, 1)$ .  
Direction vector:  $\vec{AB} = (4 - 1, 3 - 0, 1 - 2) = (3, 3, -1)$ . Vector equation:  $\mathbf{r} = (1, 0, 2) + t(3, 3, -1)$ .

### Determining if two lines intersect

- 3 Determine whether the lines  $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$  and  $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$  intersect, and if so, find the point of intersection.  
Equate components:  $1 + s = 2t$ ,  $2 - s = 3 + t$ ,  $2s = 1 - t$ . From the second:  $s = -1 - t$ . Substitute into the first:  $1 + (-1 - t) = 2t$ , so  $-t = 2t$ , giving  $t = 0$ , then  $s = -1$ . Check the third equation:  $2(-1) = 1 - 0$ ?  $-2 = 1$  — false. The lines do not intersect (they are skew, since they are not parallel either).

### The equation of a plane

- 4 Find the Cartesian equation of the plane with normal vector  $(2, -1, 3)$  passing through the point  $(1, 2, -1)$ .  
The plane has the form  $2x - y + 3z = d$ . Substituting the point:  $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$ .  
Equation:  $2x - y + 3z = -3$ .
- 5 Find the Cartesian equation of the plane containing the points  $A(1, 0, 0)$ ,  $B(0, 1, 0)$ , and  $C(0, 0, 1)$ .  
 $\vec{AB} = (-1, 1, 0)$ ,  $\vec{AC} = (-1, 0, 1)$ . Normal  
 $= \vec{AB} \times \vec{AC} = (1 \cdot 1 - 0 \cdot 0, 0 \cdot (-1) - (-1) \cdot 1, (-1) \cdot 0 - 1 \cdot (-1)) = (1, 1, 1)$ . Plane:  $x + y + z = d$ ;  
substituting A:  $1 + 0 + 0 = 1$ . Equation:  $x + y + z = 1$ .

### Angle between a line and a plane

- 6 Find the angle between the line with direction  $(1, 2, 2)$  and the plane  $x + y + z = 5$ , to the nearest degree.  
Plane normal:  $\mathbf{n} = (1, 1, 1)$ .  $\mathbf{d} \cdot \mathbf{n} = 1 + 2 + 2 = 5$ .  $|\mathbf{d}| = 3$ ,  $|\mathbf{n}| = \sqrt{3}$ .  $\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{5}{3\sqrt{3}} \approx 0.962$ ,  
so  $\theta \approx 74^\circ$ .

## Distance from a point to a plane

- 7 Find the shortest distance from the point  $(3, 1, -2)$  to the plane  $2x - y + 2z = 6$ , using

$$d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$$
$$d = \frac{|2(3) - 1(1) + 2(-2) - 6|}{\sqrt{4 + 1 + 4}} = \frac{|6 - 1 - 4 - 6|}{3} = \frac{|-5|}{3} = \frac{5}{3}.$$

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## Finding the angle between two planes

- 8 Find the angle between the planes  $x + 2y - z = 3$  and  $2x - y + z = 5$ , to the nearest degree.

Normals:  $\mathbf{n}_1 = (1, 2, -1)$ ,  $\mathbf{n}_2 = (2, -1, 1)$ .  $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$ .  $|\mathbf{n}_1| = \sqrt{6}$ ,  $|\mathbf{n}_2| = \sqrt{6}$ .

$$\cos \theta = \frac{|-1|}{\sqrt{6} \cdot \sqrt{6}} = \frac{1}{6}, \text{ so } \theta \approx 80^\circ.$$

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## Finding the intersection line of two planes

- 9 Find a vector equation for the line of intersection of the planes  $x + y + z = 6$  and  $x - y + 2z = 5$ , using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.

Direction:

$$\mathbf{n}_1 \times \mathbf{n}_2 = (1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1), 1 \cdot 1 - 1 \cdot 2, 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2).$$

To find a point, try  $z = 0$ :  $x + y = 6$  and  $x - y = 5$ , giving  $x = 5.5$ ,  $y = 0.5$ . Point:  $(5.5, 0.5, 0)$ . Line:

$$\mathbf{r} = (5.5, 0.5, 0) + t(3, -1, -2).$$

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## Testing whether a line lies within a plane

- 10 Determine whether the line  $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$  lies entirely within the plane  $x + 3y + z = 10$ .

A line lies within a plane if (1) its direction vector is perpendicular to the plane's normal, and (2) a point on the line satisfies the plane's equation. Normal of the plane:  $(1, 3, 1)$ . Direction of line:  $(2, -1, 1)$ . Dot product:  $1(2) + 3(-1) + 1(1) = 2 - 3 + 1 = 0$  ✓ (perpendicular to normal, so parallel to the plane). Check the point  $(1, 2, 3)$ :  $1 + 3(2) + 3 = 1 + 6 + 3 = 10$  ✓ (satisfies the plane equation). Since both conditions hold, the line lies entirely within the plane.

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## Finding the point where a line meets a plane

- 11 Find the point where the line  $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$  intersects the plane  $2x + y - z = 3$ .

Substitute the parametric coordinates  $x = 1 + t$ ,  $y = 2t$ ,  $z = 2 - t$  into the plane equation:

$$2(1 + t) + 2t - (2 - t) = 3. \text{ Expand: } 2 + 2t + 2t - 2 + t = 3, \text{ so } 5t = 3, \text{ giving } t = \frac{3}{5}.$$

Substituting back:

$$x = 1 + \frac{3}{5} = \frac{8}{5}, y = 2\left(\frac{3}{5}\right) = \frac{6}{5}, z = 2 - \frac{3}{5} = \frac{7}{5}. \text{ Point of intersection: } \left(\frac{8}{5}, \frac{6}{5}, \frac{7}{5}\right).$$

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## Shortest distance between two skew lines

- 12 Two skew lines have direction vectors  $\mathbf{d}_1 = (1, 0, 1)$  and  $\mathbf{d}_2 = (0, 1, 1)$ , and pass through the points  $A(0, 0, 0)$  and  $B(1, 1, 0)$  respectively. Find the shortest distance between the two lines, using

$$d = \frac{|\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}.$$

$$\mathbf{d}_1 \times \mathbf{d}_2 = (0 \cdot 1 - 1 \cdot 1, 1 \cdot 0 - 1 \cdot 1, 1 \cdot 1 - 0 \cdot 0) = (-1, -1, 1). \vec{AB} = (1, 1, 0).$$

$$\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) = 1(-1) + 1(-1) + 0(1) = -2. |\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{1 + 1 + 1} = \sqrt{3}. \text{ Distance} \\ = \frac{|-2|}{\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

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## Foot of the perpendicular from a point to a line

- 13 Find the foot of the perpendicular from the point  $P(4, 0, 1)$  to the line  $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$ .

A general point on the line is  $(1 + t, t, t)$ . The vector from this point to  $P$  is

$(4 - (1 + t), 0 - t, 1 - t) = (3 - t, -t, 1 - t)$ . For the foot of the perpendicular, this vector must be perpendicular to the direction  $(1, 1, 1)$ :  $(3 - t)(1) + (-t)(1) + (1 - t)(1) = 0$ , i.e.  $3 - t - t + 1 - t = 0$ , so  $4 - 3t = 0$ , giving  $t = \frac{4}{3}$ . Foot of perpendicular:  $(1 + \frac{4}{3}, \frac{4}{3}, \frac{4}{3}) = (\frac{7}{3}, \frac{4}{3}, \frac{4}{3})$ .

- 14 Determine whether the planes  $2x - y + 3z = 4$  and  $-4x + 2y - 6z = -8$  represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

The second equation's coefficients  $(-4, 2, -6)$  are exactly  $-2$  times the first equation's coefficients  $(2, -1, 3)$ , and the constant  $-8$  is also  $-2$  times  $4$ . Since the entire equation is a scalar multiple of the first, the two equations describe the same plane.

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## Checking whether a point lies on a plane

- 15 Determine whether the point  $P(2, -1, 4)$  lies on the plane  $3x + 2y - z = 1$ , and if not, state whether it lies on the same side as the origin.

Substituting  $P$ :  $3(2) + 2(-1) - 4 = 6 - 2 - 4 = 0 \neq 1$ , so  $P$  does not lie on the plane. Substituting the origin  $(0, 0, 0)$ :  $3(0) + 2(0) - 0 = 0$ . Since the plane value at  $P$  ( $0$ ) equals the plane value at the origin ( $0$ ), and both are less than the required value of  $1$ ,  $P$  lies on the same side of the plane as the origin.

- 16 Find the value of  $k$  such that the line  $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$  is parallel to the plane  $x + 2y - z = 5$ .

A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector. Normal of the plane:  $(1, 2, -1)$ . Direction of line:  $(2, -1, k)$ . Setting the dot product to zero:

$$1(2) + 2(-1) + (-1)(k) = 0, \text{ so } 2 - 2 - k = 0, \text{ giving } k = 0.$$

- 17 A surveyor models a hillside as the plane  $2x + 3y + 6z = 42$ , where  $x$  and  $y$  are horizontal coordinates (in metres) and  $z$  is height (in metres). Find the height  $z$  of the hillside at the point where  $x = 6$  and  $y = 4$ , and find the plane's intercept on the  $z$ -axis (i.e. the height at  $x = 0, y = 0$ ).

Substituting  $x = 6, y = 4$ :  $2(6) + 3(4) + 6z = 42$ , so  $12 + 12 + 6z = 42$ , giving  $6z = 18$ , so  $z = 3$  metres. The  $z$ -intercept (at  $x = 0, y = 0$ ):  $6z = 42$ , so  $z = 7$  metres.

- 18 This question has three parts. Two lines are given by  $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$  and  $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$ . (a) Show that both lines lie in the plane  $z = 1$ . (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.
- (a) Both lines have  $z$ -component fixed at 1 (since the direction vectors  $(1, 1, 0)$  and  $(1, -1, 0)$  have zero  $z$ -component, and both starting points have  $z = 1$ ), so every point on either line satisfies  $z = 1$ ; both lines lie in the plane  $z = 1$ . (b) Equating:  $2 + s = t$  and  $s = 2 - t$ . Substituting the second into the first:  $2 + (2 - t) = t$ , so  $4 - t = t$ , giving  $t = 2$ , then  $s = 0$ . Point:  $\mathbf{r}_1$  at  $s = 0$  gives  $(2, 0, 1)$ ; checking  $\mathbf{r}_2$  at  $t = 2$ :  $(0, 2, 1) + 2(1, -1, 0) = (2, 0, 1)$  ✓. Intersection point:  $(2, 0, 1)$ . (c) Direction vectors:  $(1, 1, 0)$  and  $(1, -1, 0)$ . Dot product:  $1 - 1 + 0 = 0$ . Since the dot product is zero, the lines are perpendicular, so the angle between them is  $90^\circ$ .